

Sri Lanka Institute of Information Technology

Human Computer Interaction (IT3060)

Lab Exercise 06

FitFlow — Release Preparation and Internal Testing



Details

Campus: Malabe Campus

Name: Jayawardhana O K T C

Student ID: IT23230774

Year: 3rd Year, 2nd Semester

Group: WE 1.1

Date: 04/10/2026

Contents

Overview

Activity 1 – Generate Signed APK/AAB

Activity 2 – Prepare App Icons, Screenshots and Store Assets

Activity 3 – Configure Google Play Console

Activity 4 – Configure App Store Connect and TestFlight

Activity 5 – Prepare a Privacy Policy and Release Notes

Activity 6 – Perform Internal Testing Deployment

Appendix – Build and test commands

GitHub repository: <https://github.com/CHANUKA-01/fitflow-redesign>

All documents referenced in this report are in the repository under docs/lab06/, the store assets under store-assets/ and the app under frontend/mobile/.

Overview

Lab 05 selected Flutter for the FitFlow mobile client. For this lab the Flutter app was built ("Focus Flow" design from Lab 03), signed for release, and prepared for both stores. Every screen implements a requirement from earlier labs (R01–R16, NFR1–NFR5, usability issues U01–U14).

Item	Value
App	FitFlow: Workouts & Nutrition — package / bundle ID io.fitflow.app
Version	1.1.0 (build / versionCode 2)
Framework	Flutter 3.47.6, Dart 3.13.5; Android 7.0+ (API 24–36), iOS 15+
Features	Daily AI-style plan on Home, explained planner, one-tap set logging, on-device meal photo review, nutrition, private circles, progress, privacy controls
Store accounts	Not available (Google Play US\$25, Apple US\$99) — Activities 3 and 4 are fully prepared configurations; the signed builds and all assets are real

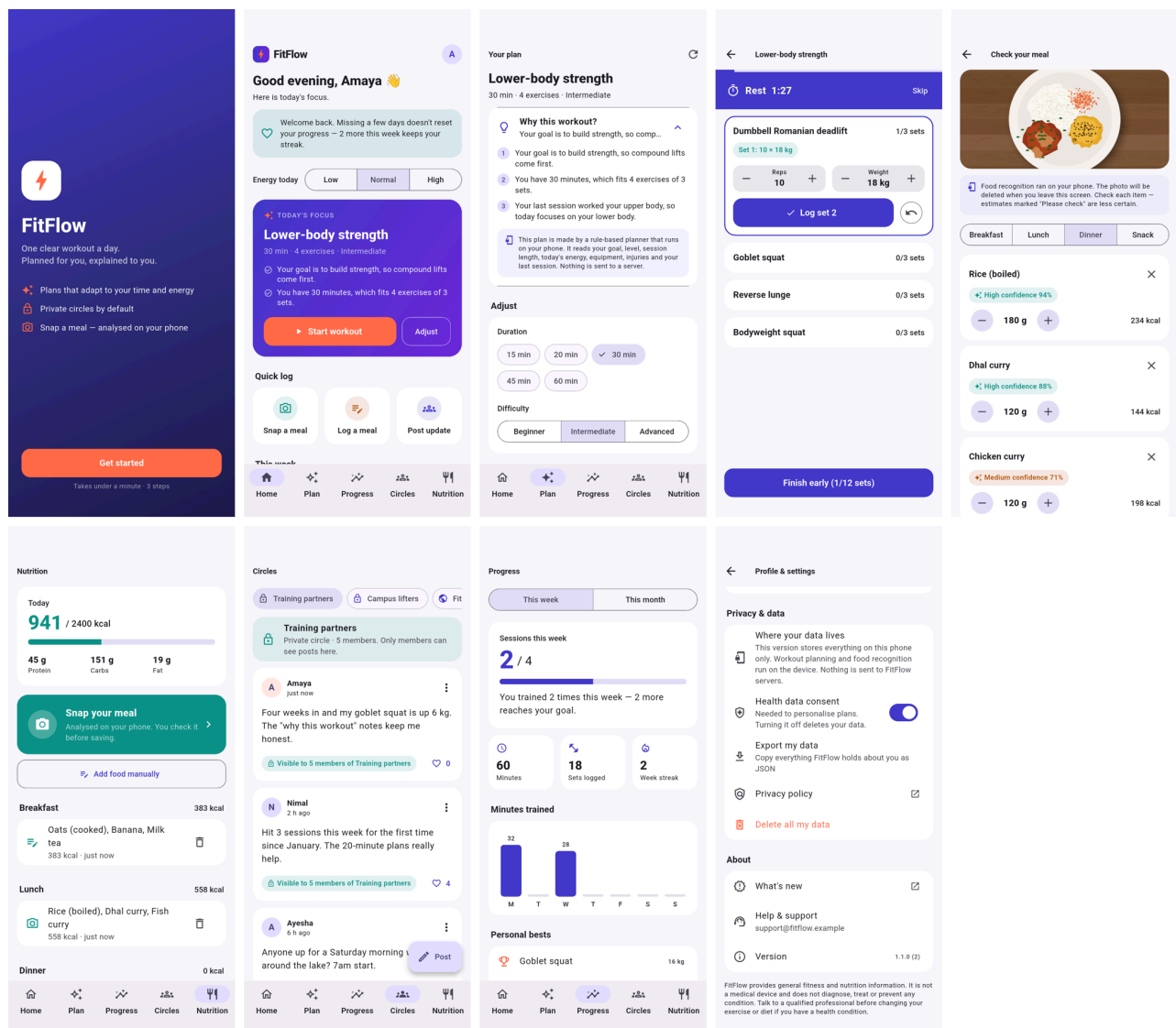


Figure 1 – FitFlow 1.1.0 running on the Pixel 9a emulator (signed build, demo data): welcome, home, plan, logger, meal review, nutrition, circles, progress, privacy.

Activity 1 – Generate Signed APK/AAB

1.1 Build details

Item	Value
Application ID	io.fitflow.app
Version name / code	1.1.0 / 2 (baseline 1.0.0 / 1)
Min / target / compile SDK	24 (Android 7.0) / 36 (Android 16) / 36
ABIs	arm64-v8a, armeabi-v7a, x86_64
Signing key	Upload key fitflow-upload, RSA 4096, SHA384withRSA, PKCS12, valid to 2056
Optimisation	R8 minify + resource shrinking; Dart obfuscation with symbols split out

1.2 Release configuration (android/app/build.gradle.kts)

```
signingConfigs {
    if (releaseStoreFile != null) {
        create("release") {
            storeFile = file(releaseStoreFile) // from key.properties or FITFLOW_* env vars
            storePassword = signingValue("storePassword", "FITFLOW_KEYSTORE_PASSWORD")
            keyAlias = signingValue("keyAlias", "FITFLOW_KEY_ALIAS")
            keyPassword = signingValue("keyPassword", "FITFLOW_KEY_PASSWORD")
            storeType = "pkcs12"
        }
    }
}
buildTypes {
    release {
        signingConfig = signingConfigs.findByName("release") // no fallback to the debug key
        isMinifyEnabled = true
        isShrinkResources = true
        proguardFiles(getDefaultProguardFile("proguard-android-optimize.txt"), "proguard-rules.pro")
    }
}
```

- The template signed release builds with the **debug key**; that was removed. A Gradle task-graph check stops any release build with a clear error if the upload key is missing, while debug builds still work.
- Android 11+ <queries> added so the privacy-policy and support links open on modern phones.

1.3 Keystore and signing

```
keytool -genkeypair -v -storetype PKCS12 \
-keystore ~/.fitflow-signing/fitflow-upload.jks -alias fitflow-upload \
-keyalg RSA -keysize 4096 -validity 10950 \
-dname "CN=FitFlow Upload Key, OU=IT3060 WE 1.1, O=FitFlow, L=Malabe, ST=Western, C=LK"
```

Secret	Location	In git?
Keystore file	~/.fitflow-signing/fitflow-upload.jks (outside repo, mode 600)	No
Passwords + alias	~/.fitflow-signing/key.properties; android/key.properties is a symlink	No (git-ignored)
CI copies	GitHub Actions secrets (keystore base64 + passwords)	No

- **Play App Signing:** Google holds the app signing key; the team holds only the upload key, which can be reset through Play Console if lost.

- **Backup:** two encrypted offline copies (USB + password manager); password only in the password manager; access limited to release owner and one deputy; restore test each semester.
- **Incident found and fixed:** the project is on an iCloud-synced Desktop, so key.properties (passwords) was being uploaded to iCloud. It was moved outside the synced folder.
- Upload certificate SHA-256:
93:B8:CF:64:31:72:92:77:8C:DA:A9:3D:ED:CB:96:90:67:EB:08:FA:3C:78:4D:4E:93:5D:29:8B:38:BA:57:51

1.4 Build commands

```
cd frontend/mobile
scripts/release_android.sh          # clean, analyse, test, build AAB + APKs, verify, write evidence
# equivalent steps:
flutter build appbundle --release --obfuscate --split-debug-info=build/debug-info
flutter build apk --release --obfuscate --split-debug-info=build/debug-info
flutter build apk --release --split-per-abi --obfuscate --split-debug-info=build/debug-info
```

1.5 Verification

Check	Tool	Result
AAB signature	jarsigner -verify	jar verified — CN=FitFlow Upload Key
APK signature	apksigner verify	Verifies (APK Signature Scheme v2), RSA 4096
Signer = keystore	SHA-256 compare	Match (93b8cf64...5751)
Code shrinking	R8 mapping.txt	Present — code obfuscated
Play download size	bundletool get-size	≈ 8.0 MB (arm64) vs 48.6 MB AAB upload file
Permissions	aapt2 dump badging	None shown to the user (no INTERNET, CAMERA or storage)
Install test	bundletool install-apks	Installed and run on Pixel 9a emulator (Android 16)
Start-up (NFR3)	am start -W	3.8–7.5 s on emulator; control app 2.5–5.5 s → emulator not valid; physical device pending

Excerpt of the generated evidence log (docs/lab06/evidence/release-evidence-1.1.0+2.txt):

FitFlow Android release evidence

Version (pubspec): 1.1.0+2

Generated: 2026-10-04 00:52:23 UTC

Commit: 9fd3e05

Flutter: Flutter 3.47.6 • channel stable • <https://github.com/flutter/flutter.git>

== Artefacts ==

48580237 build/app/outputs/bundle/release/app-release.aab
17624050 build/app/outputs/flutter-apk/app-arm64-v8a-release.apk
14868924 build/app/outputs/flutter-apk/app-armeabi-v7a-release.apk
49516995 build/app/outputs/flutter-apk/app-release.apk
19124209 build/app/outputs/flutter-apk/app-x86_64-release.apk

== SHA-256 checksums ==

eb414385335482a18294aadf4db7d8311c9e0d5a7a150d1958e1a02542949c build/app/outputs/bundle/release/app-release.aab
bd1a181298c0b47a379a72f5e09f35196aba78f66e2a55f0fca349e8e0e89a33 build/app/outputs/flutter-apk/app-release.apk

== AAB signature (jarsigner) ==

jar verified.

Owner: CN=FitFlow Upload Key, OU=IT3060 WE 1.1, O=FitFlow, L=Malabe, ST=Western, C=LK

Valid from: Sat Oct 03 21:23:08 IST 2026 until: Mon Sep 25 21:23:08 IST 2056

SHA256:

93:B8:CF:64:31:72:92:77:8C:DA:A9:3D:ED:CB:96:90:67:EB:08:FA:3C:78:4D:4E:93:5D:29:8B:38:BA:57:51

Signature algorithm name: SHA384withRSA

== APK signature (apksigner) ==

Verifies

Verified using v1 scheme (JAR signing): false

Verified using v2 scheme (APK Signature Scheme v2): true

Verified using v3 scheme (APK Signature Scheme v3): false

Verified using v3.1 scheme (APK Signature Scheme v3.1): false

Verified using v4 scheme (APK Signature Scheme v4): false

Signer #1 certificate DN: CN=FitFlow Upload Key, OU=IT3060 WE 1.1, O=FitFlow, L=Malabe, ST=Western, C=LK

Signer #1 certificate SHA-256 digest: 93b8cf64317292778cdaa93dedcb969067eb08fa3c784d4e935d298b38ba5751

Signer #1 key size (bits): 4096

== Manifest (aapt2) ==

package: name='io.fitflow.app' versionCode='2' versionName='1.1.0'

minSdkVersion:'24'

targetSdkVersion:'36'

uses-permission: name='io.fitflow.app.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION'

Activity 2 – Prepare App Icons, Screenshots and Store Assets

All assets are generated by code (Python/Pillow) and screenshots are captured from the real app by an automated Flutter integration test, so they can be regenerated after any change. A validator checks every file against Google Play and App Store rules: **72 of 72 checks passed**.

2.1 Icon and branding



Figure 2 – App Store/Play icon, Android adaptive icon, Android 13 themed (monochrome) icon.

Asset	Size / format	Rule
Play hi-res icon	512×512 PNG	Full square, Google applies mask
App Store icon	1024×1024 PNG, no alpha	Apple rejects transparency
Adaptive icon	108 dp layers, all densities	Mark inside 66 dp safe zone
Themed icon	Monochrome layer	Android 13+ tints it
iOS Applcon set	20–1024 pt @1x–@3x	Generated by flutter_launcher_icons
Splash	Brand indigo, Android 12+ and iOS	No white flash at start

2.2 Feature graphic and banners

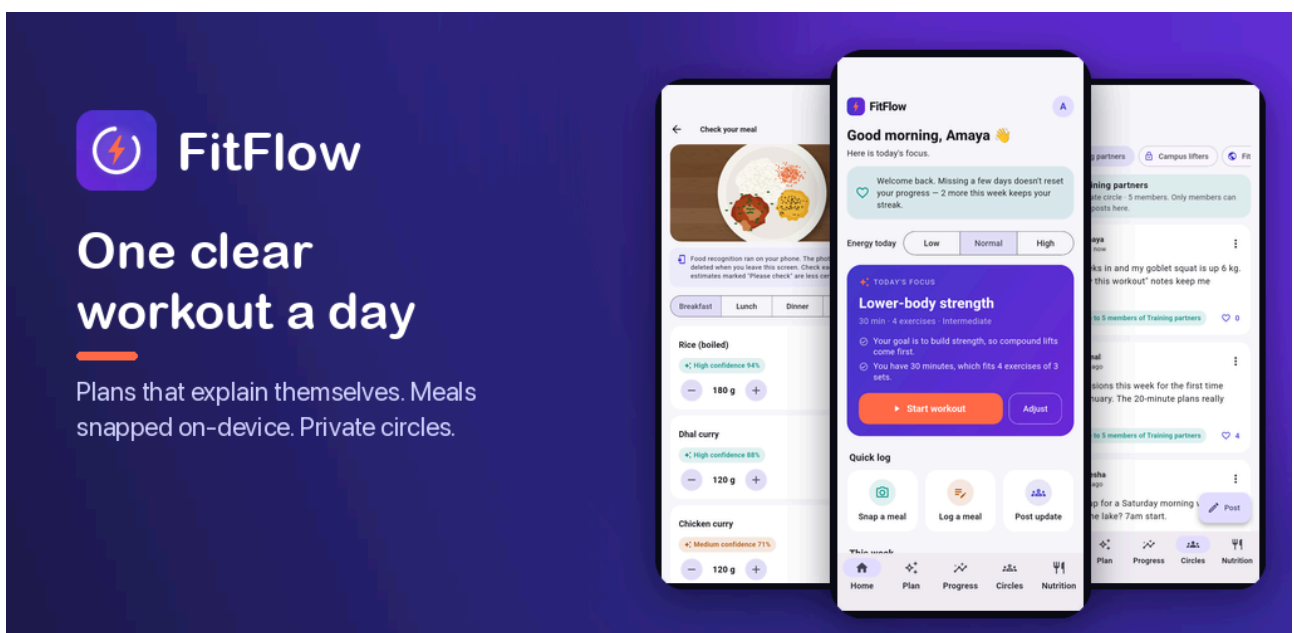


Figure 3 – Google Play feature graphic (1024×500).

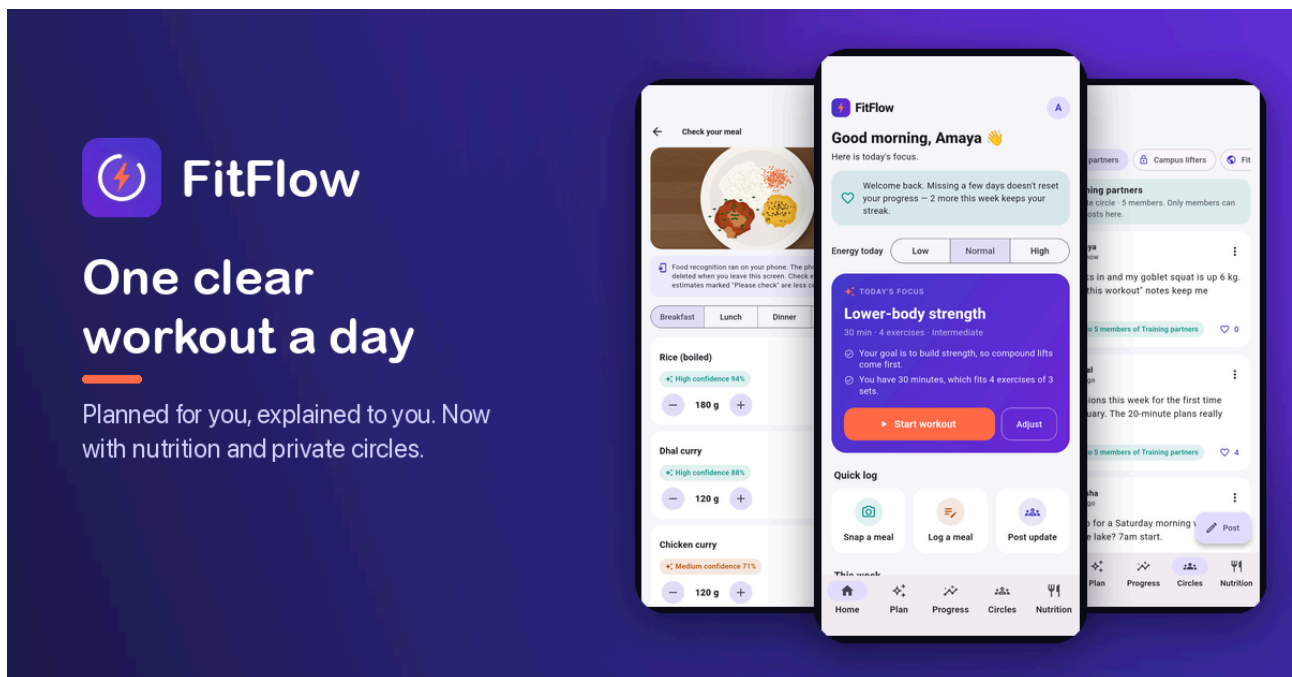


Figure 4 – Promotional banner (1200×628).

2.3 Screenshots

Store slot	Size	Count
Play – phone	1080×1920	8
Play – 7-inch tablet	1200×1920 portrait + 1920×1200 landscape	8 + 8
Play – 10-inch tablet	1600×2560 portrait + 2560×1600 landscape	8 + 8
App Store – iPhone 6.9"	1320×2868	8
App Store – iPad 13"	2064×2752	8

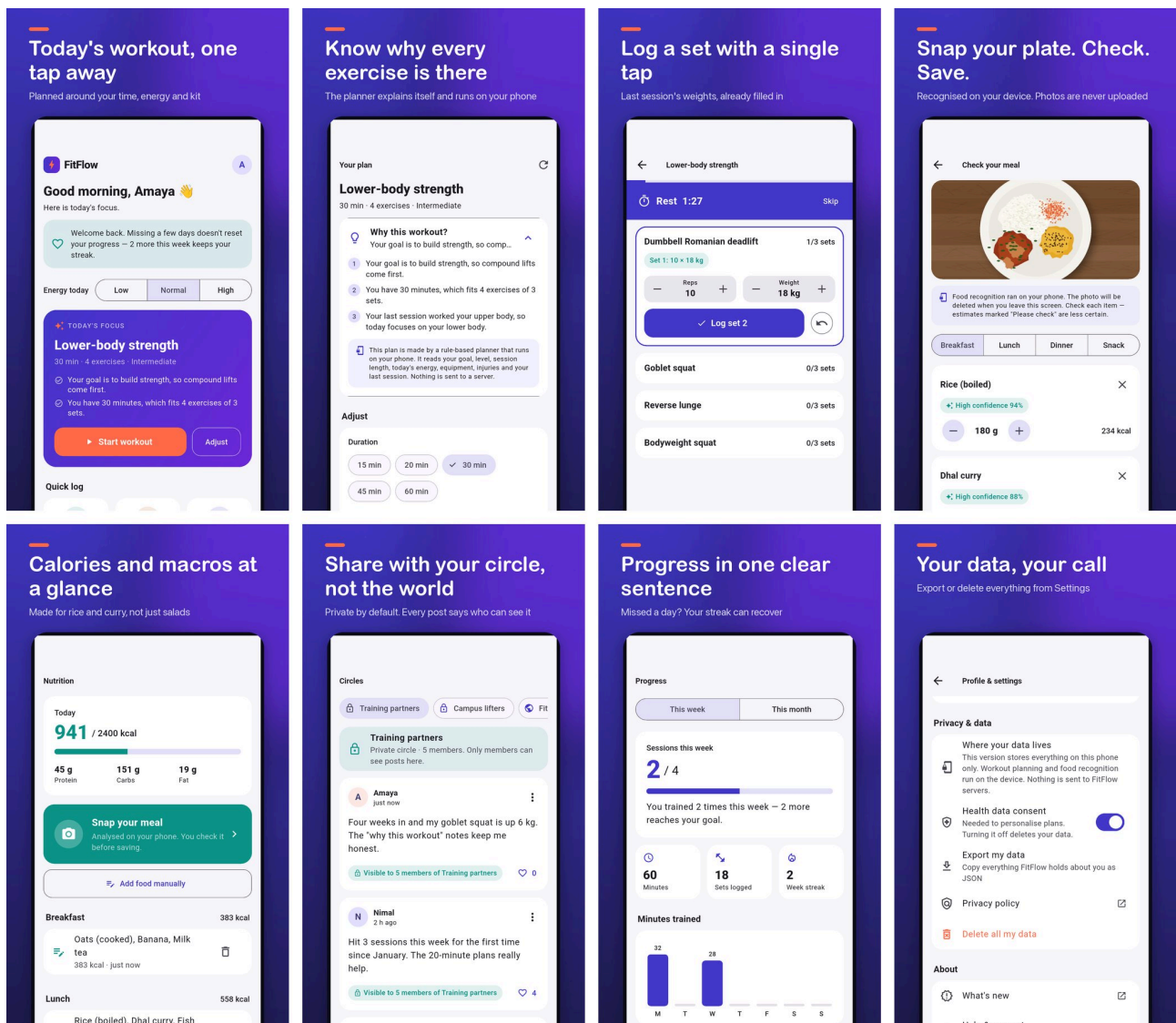


Figure 5 – Google Play phone screenshots 01–08 (captioned, in upload order).

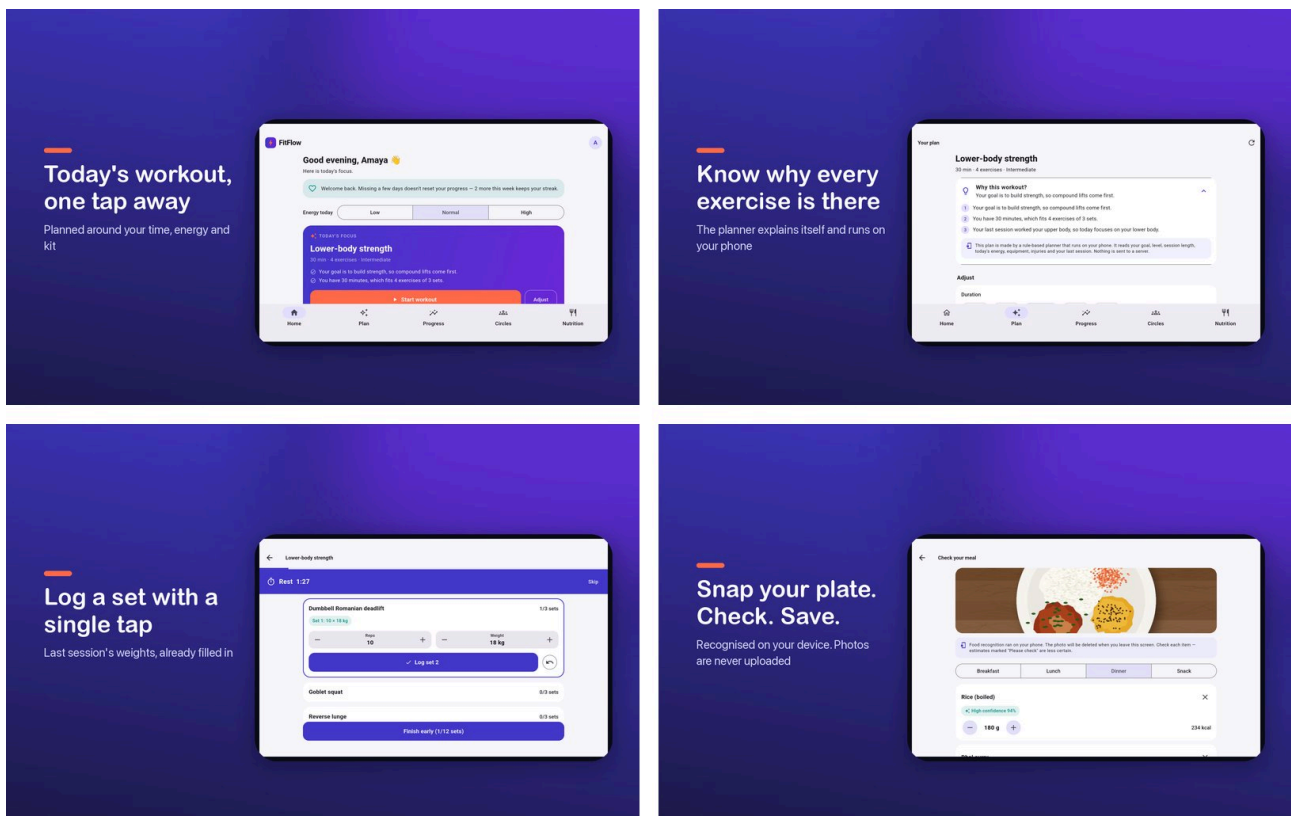


Figure 6 – 7-inch tablet, landscape.

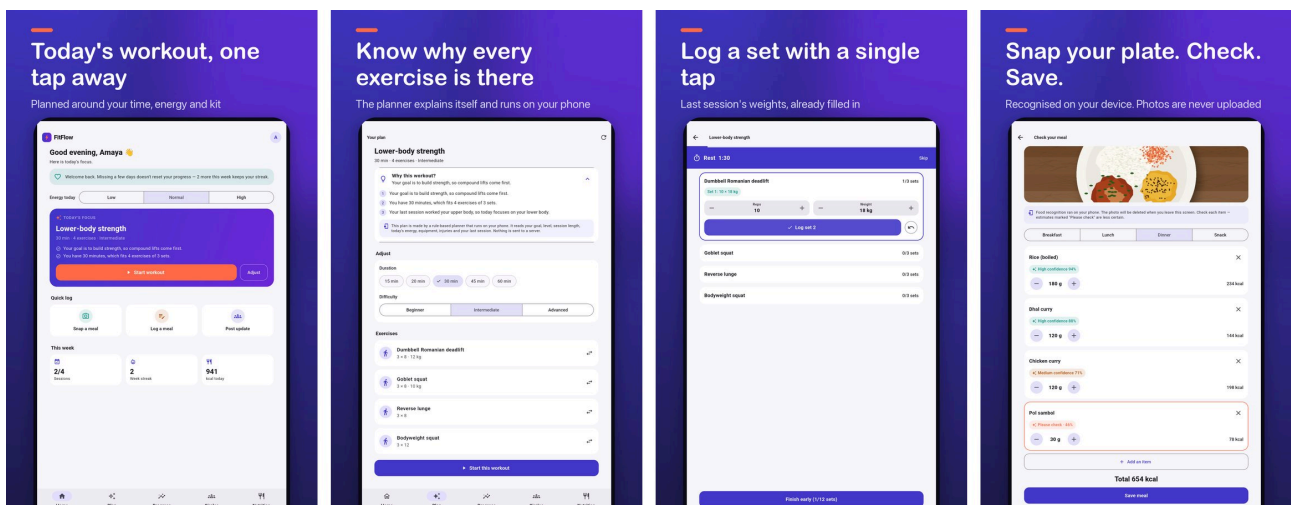


Figure 7 – 10-inch tablet, portrait.

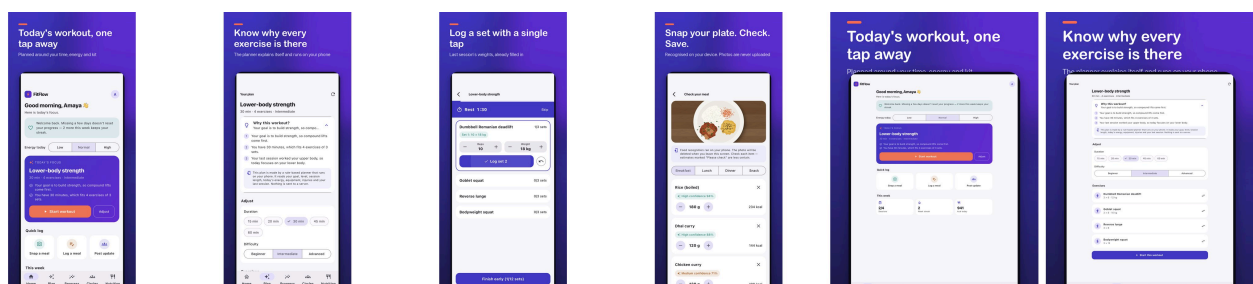


Figure 8 – App Store: iPhone 6.9" (first four) and iPad 13".

#	Screen	Caption	Requirement
01	Home	Today's workout, one tap away	R01, R14, U04

02	Plan	Know why every exercise is there	R03–R05, R15
03	Logger	Log a set with a single tap	R06, R07, NFR1
04	Meal review	Snap your plate. Check. Save.	R08, R09, U08
05	Nutrition	Calories and macros at a glance	R08
06	Circles	Share with your circle, not the world	R11, U01
07	Progress	Progress in one clear sentence	R10, R12, R13
08	Privacy	Your data, your call	NFR5

2.4 Video, listing text and folder structure

- **App Store app preview:** 886×1920, H.264, 30 fps, silent stereo AAC, 23 s of real app footage recorded on the iPhone simulator with a clean 9:41 status bar (Apple requires real footage, 15–30 s).
- **Play promo video:** 1080×1920, 27 s — uploaded to YouTube (unlisted) and linked in the listing.
- **Listing text:** title 29/30, Play short description 78/80, App Store subtitle 24/30, keywords 98/100, promotional text 144/170, description ≈2 360/4000.
- **Honest metadata:** the planner is rule-based in this prototype, so the listing says "smart planner" not "AI" (Apple 2.3 / Play misleading claims). An unverified WCAG claim was removed.

```
store-assets/
├─ android/  icon/ feature-graphic/ screenshots/{phone, tablet-7in, tablet-10in}
├─ ios/      icon/ screenshots/{iphone-6.9in, ipad-13in} previews/
├─ promo/    banners/ video/
├─ listing/  google-play/ app-store/
└─ _tools/   generators + validate_assets.py
Naming: ff_<platform>_<device>[-<orientation>]_<NN>_<screen>_<W>x<H>.png
e.g. ff_android_phone_01_home_1080x1920.png
```

2.5 Problems found and fixed

Problem	Fix
Circle audience label truncated ("Visible to 5 mem...") — undid the U01 privacy fix	Label given its own space, wraps
Tablet layouts stretched across 2560 px	Content capped to a centred 720 dp column
Raw phone screenshot 2.24:1 — Play rejects > 2:1	Framed onto 9:16 canvas with caption
iOS build: codesign rejected files tagged by iCloud	Build output moved outside synced Desktop

Activity 3 – Configure Google Play Console

Status: fully prepared; not submitted because no Google Play developer account was available. Full configuration: docs/lab06/03-google-play-console.md.

Section	Configuration
App	FitFlow: Workouts & Nutrition · en-GB · App · Free
Category	Health & Fitness; tags Workout, Fitness tracking, Nutrition
Store listing	Title, short and full description, 512 icon, feature graphic, 8 phone + 16 + 16 tablet screenshots, YouTube promo
Privacy policy	https://chanuka-01.github.io/fitflow-redesign/privacy-policy/
Content rating (IARC)	No violence/sex/gambling/drugs; users interact = Yes (circles, with report/delete) → expected PEGI 3 / Everyone
Target audience	18+ (16–17 allowed); not designed for children
Ads / Advertising ID	No ads; Advertising ID not used
Data safety	No data collected or shared (all on device, no INTERNET permission); in-app deletion available. Planned answers prepared for the backend release
Health apps declaration	Activity & fitness tracking; nutrition; no medical/regulated features
Play App Signing	Google-generated app signing key; upload certificate SHA-256 93:B8:CF...:57:51
Testing tracks	Internal (team, ≤100) → Closed (Google Group) → Production. Personal accounts need 12 testers × 14 days of closed testing
Release	Internal: upload app-release.aab (1.1.0, code 2) + mapping.txt; notes from Activity 5
Countries	Sri Lanka first; then India, UK, Australia, Singapore
Device exclusions	None needed — minSdk 24 excludes old Android; camera not required

3.1 Pre-launch report

Play runs every testing-track upload on real devices. Expected: no crashes (none seen in 7 device configurations), possible accessibility suggestions (contrast audit is open), no security warnings (no cleartext traffic, no exported components except the launcher). Errors block promotion; warnings go to the bug log.

3.2 Store listing experiments (A/B tests)

#	Hypothesis	Variant	Metric
E1	Leading with the explained plan converts better	Screenshot order 02-01-03	Retained installers (1 day)
E2	Privacy-first short description attracts users who stay	"Private fitness planner: workouts that explain themselves, data stays on your phone."	Retained installers
E3	Food recognition in the feature graphic lifts nutrition searches	Meal-review phone in centre	First-time installers

50/50 split, at least 7 days, apply only at 90 % confidence; E1 first.

Activity 4 – Configure App Store Connect and TestFlight

Status: configuration prepared and the iOS app built and run on the iPhone 18 Pro Max and iPad Pro 13-inch simulators (iOS 27, Xcode 27); not uploaded because no Apple Developer Program membership was available. Full configuration: docs/lab06/04-app-store-connect-testflight.md.

Section	Configuration
App record	iOS · FitFlow: Workouts & Nutrition · English (U.K.) · bundle io.fitflow.app · SKU FITFLOW-IOS-001
Subtitle / category	"Workouts planned for you" · Health & Fitness (secondary Food & Drink)
Keywords (98/100)	workout,planner,gym,fitness,strength,nutrition,calorie,meal,tracker,dumbbell,exercise,habit,streak
Age rating	4+ questionnaire; user-generated content = Yes with report/delete
Privacy nutrition label	"Data Not Collected" for 1.1.0 (no network access). Planned label prepared for backend release (Fitness, Name/Email, User content, Diagnostics — none used for tracking)
Screenshots / preview	8 × iPhone 6.9" (1320×2868), 8 × iPad 13" (2064×2752), 23 s app preview
Export compliance	ITSAAppUsesNonExemptEncryption = NO added to Info.plist
Upload	flutter build ipa → Xcode Organizer or xcrun altool

4.1 TestFlight

Group	Type	Members	Review
FitFlow Team	Internal	≤100 App Store Connect users	None
FitFlow Beta – Campus	External	20–30 SLIIT students who train	Beta App Review (first build)
FitFlow Beta – Public link	External	Link capped at 100	Beta App Review

- Builds expire after **90 days**: new build at least every 60 days or after any critical fix; superseded builds expired manually; build number +1 per upload.
- "What to test": onboarding → start workout → log 3 sets → snap a meal and correct an item → post to private circle and check the audience label → send feedback with screenshot.

4.2 Apple review compliance (health)

Guideline	FitFlow
1.4.1 Physical harm	Reasons shown for every plan; conservative volume; professional-advice disclaimer
1.2 User-generated content	Report + delete on posts; private by default; contact in listing (block-user logged as gap)
2.3 Accurate metadata	Real screenshots; "smart planner", not "AI"
HealthKit / 5.1.3	Not used in 1.1.0; rules documented if added (no ads, no iCloud storage)
5.1.1 Data	Privacy policy in app and listing; explicit consent; in-app deletion
Camera purpose strings	Explain on-device analysis and no upload

Activity 5 – Prepare a Privacy Policy and Release Notes

Full texts: docs/lab06/05-privacy-policy.md and docs/lab06/05-release-notes.md. The policy is linked in the app (onboarding consent step and Settings → Privacy policy) and in both store listings at <https://chanuka-01.github.io/fitflow-redesign/privacy-policy/> — publishing it with GitHub Pages is the remaining step.

5.1 Privacy policy (summary of sections)

Section	Content
Summary	Everything stays on the phone; no accounts, ads or tracking; photos analysed on device then deleted
Data used	Profile, health/fitness (injuries, workouts), nutrition, meal photos, posts, consent record; no contacts, location, ad ID
Health data & consent	Special-category data (GDPR Art. 9; Sri Lanka PDPA No. 9 of 2022) — explicit consent at onboarding, withdrawable in Settings
AI processing	Rule-based on-device planner with reasons; on-device food recognition with confidence; suggestions only, no Art. 22 decisions
Social features	Private circle by default; audience shown on every post; delete/report; local-only in 1.1.0
Storage & security	App-private storage, device encryption; removed on uninstall or delete
Sharing	None; future providers (e.g. Firebase Crashlytics) listed before they are added
Rights	Access/portability (Export), correction, erasure (Delete all), withdraw consent, complaint to authority
Children	16+; not directed at children; no under-13 data
HIPAA	Not a covered entity, so HIPAA does not apply; HIPAA-style safeguards followed
Disclaimer, changes, contact	Not a medical device; versioned changes with in-app notice; contact support@fitflow.example (placeholder)

5.2 Release notes – Google Play (449/500 characters)

FitFlow has been redesigned around one clear workout a day.

- Today's plan waits on your home screen. Start it in one tap.
- See why every exercise is there, then shorten it, swap moves or regenerate.
- Log a set with a single tap: last time's weights are already filled in.
- Snap your meal: food is recognised on your phone and you check it before saving.
- Private circles: every post shows exactly who can see it.

Your data stays on your device.

5.3 Update history and known issues

Version	Build	Date	Summary
1.1.0	2	2026-10-04	Focus Flow redesign
1.0.0	1	—	Template baseline, never distributed

- Food recognition uses a demonstration model — results must be checked.
- Circles are stored on the device only.
- Start-up target (2 s) still to be measured on a physical device.

Activity 6 – Perform Internal Testing Deployment

Only observed results are reported; items not yet done are marked Pending. Full record: docs/lab06/06-internal-testing.md.

6.1 Deployment

Channel	Status
Google Play Internal Testing	Prepared (Activity 3) — needs Play account
Apple TestFlight	Prepared (Activity 4) — needs Apple Developer account
Signed build direct install	Done — AAB → bundletool install-apks on Pixel 9a emulator
Firebase App Distribution	Planned free alternative using the signed universal APK

6.2 Results

Test	Result
flutter analyze	No issues
Unit/widget tests	5/5 passed (injury rules, equipment, reasons, low-energy volume, onboarding consent)
Integration test (9 screens)	Passed on 7 configurations: phone, 7"/10" tablets portrait + landscape, iPhone 6.9", iPad 13"
Manual walkthrough T01–T09	All passed (T08 after fixing B01)
Crashes	None in any run (logcat checked)
Start-up (NFR3)	Emulator not valid (control app as slow); physical device Pending
Battery	Pending — physical device needed
Crashlytics	Not integrated yet — needs INTERNET permission, so planned with privacy-policy update

6.3 Bug log

ID	Severity	Description	Fix	Status
B01	Major	Audience label truncated (U01 regression)	Own space, wraps	Fixed
B02	Minor	Chip checkmark over lock icon	showCheckmark: false	Fixed
B03	Major	Tablet content stretched	720 dp column	Fixed
B04	Major	Links fail on Android 11+	Manifest <queries>	Fixed
B05	Critical	iCloud duplicate files broke Android build	Build out of synced folder	Fixed
B06	Critical	iOS codesign rejected iCloud-tagged files	Same as B05	Fixed
B07	Major	Signing passwords in iCloud-synced folder	Moved, symlinked	Fixed
B08	Minor	Screenshot captured stale frame	Settle + precache	Fixed
B09	Minor	Emulator slept mid-test	Stay-awake in script	Fixed

6.4 Tester feedback (to be completed)

Tester	Device / OS	Tasks completed	Rating 1–5	Comment

6.5 Approval record

Criteria: no open Critical/Major bugs ✓ · tests pass ✓ · release build verified ✓ · physical-device start-up ≤ 2 s ☐ · tester feedback reviewed ☐

Role	Name	Decision	Signature	Date
Developer	Jayawardhana O K T C (IT23230774)			
Reviewer / team lead				
Lecturer				

Appendix – Build and test commands

```
# app
cd frontend/mobile
flutter pub get
flutter analyze && flutter test
flutter run                                # debug on emulator/simulator

# signed release + verification evidence
scripts/release_android.sh

# store screenshots (all devices) and preview recordings
RECORD=1 scripts/capture_screenshots.sh

# store assets (from repository root)
PY=store-assets/_tools/.venv/bin/python
$PY store-assets/_tools/make_icons.py
$PY store-assets/_tools/frame_screenshots.py
$PY store-assets/_tools/make_promo.py
store-assets/_tools/make_videos.sh
$PY store-assets/_tools/validate_assets.py
```

GitHub repository: <https://github.com/CHANUKA-01/fitflow-redesign>